

Ms2 Bacteriophages and Ampicillin on E. coli In Computational Simulations

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Introduction

Antibiotic resistance is an issue in not just developing countries, but also developed countries. *Escherichia coli* in particular develops antibiotic resistance rapidly, most prevalently through horizontal gene transfer, also known as conjugation (Nasrollahian et al., 2024). The extensive overuse of over-the-counter antibiotics within the world has contributed to the large increase in antibiotic resistance, done often without veterinarian approval (Adhikari, Lippmann, Parajuli, 2024). This is done through many mechanisms, such as efflux pumps, extended spectrum beta-lactamases, and carbapenemases. The rising trend of *E. coli* infections within leafy greens poses a legitimate problem, as *E. coli* is the most common pathogen associated with leafy green infections, causing diarrhea, hemorrhagic colitis, and other infections (Luna-Guevara et al., 2019). In 2018-2020, multiple infections of OH157 *E. coli* yielded patterns that led the United States Food and Drug Administration to find a recurring pattern, that issues with adjacent activities like cattle feces, migrating birds, and other activities often resulted in cross-contamination, as well as the geographical location of the farms (Lacombe et al., 2022). Interestingly, the majority of *E. coli* outbreaks were a result of *E. coli* OH157 producing a potent toxin, Shiga Toxin 2, as well as having an extraordinarily small infectious dose, 1-100 viable cells, and often causing complications during treatment (Centers for Disease Control, 2026 & LeBlanc, 2003). Shiga toxin 2 expressing bacteria also proliferates within many vectors, including milk (cow), beef, vegetables, flour, pork, nuts, sausage, chicken, and most animal-based food products (Wang et al., 2024). Therefore, a solution must be created to remove a large portion of an *E. coli* population to the extent where rapid reproduction of *E. coli* cells are hindered or removed, and a part of this solution must be to address antibiotic resistance. With the increase of testing on humans, using bacteriophages, many clinical trials have been successful, though the number of

formal clinical trials remains closed from strict human testing limitations, including the mass of research in Russian and Polish journals that are difficult to access (Romero-Calle et al., 2019). One field that has been explored for this solution includes the usage of bacteriophages on plant diseases, which was not recommended due to ultraviolet light, heat, and pH concerns (Jones et al., 2012). However, a solution to this issue is the application of bacteriophages on the surface of plants is for the prevention of phytopathogenic bacteria, through a spray or surface compound (Korniienko et al., 2022). Therefore, applications of bacteriophage therapies onto plant surfaces through a spray or surface compound presents a convincing solution to the wastewater and splash-based contamination methods of *E. coli* in lettuce, bean sprouts, cantaloupes, and many other produce (Hilborn et al., 1999 & Ackers et al., 1998). In more recent years, the majority of these issues were found in premade salads and Romaine Lettuce as a result of bacteria infiltrating cuts and bruises within the lettuce, even more so while washing lettuce plants (Uhlig et al., 2017). Specifically, Shiga toxin-producing *E. coli*, Norovirus, *Campylobacter jejuni*, and nontyphoidal *Salmonella* make up the highest number of illnesses and costs in leafy green outbreaks (Scharff & Yang, 2024). Furthermore, Scharff & Yang found that 75.7% of leafy green foodborne illnesses are a result of Romaine lettuce being infected. Notably, the location of various farms within Yuma County played large roles within the infection of plants (United States Food and Drug Administration, 2021). A solution including facets for antibiotic resistance, cost, prevention of cross-transmission, and possible sprays for bacteriophage applications onto plants can resolve an issue of plant-based transmissions of viral bacteria, contribute to the gap of knowledge on antibiotic-bacteriophage synergies, and contribute to the application of bacteriophages to target non-phytogenic bacteria on plants.

Literature Review

F plasmid Mechanics and Architecture

F plasmids are 100-kbp circular conjugative plasmids and are vectors for horizontal gene transfer and gene recombination. Currently, more than 200 F-like plasmids exist, and in the 60 kbp backbone, DNA transfer genes are the most conserved apart and occupy the majority of the backbone (Koraimann, 2018). F-like plasmids seem to be restricted to closely related enterobacterial genera, meaning the F plasmid is an attractive target for a generalized bacteriophage target. Interestingly, F plasmids are often found in commensal and pathogenic *E. coli* (Koraimann, 2018). The original F plasmid is constitutively derepressed because the *finO* gene is disrupted by an insertion sequence IS3 (Cheah et al., 1987). The *finO* gene encodes a 22 kDa fertility inhibition protein (FinO) in *Escherichia coli* and other Gram-negative bacteria, aiding the antisense RNA FinP in binding and silencing the *traJ* gene, which prevents the transfer of F-like plasmids between bacteria (Biesen et al., 1994; Sandercock et al., 1994). This means bacteria with a F-like plasmid will perpetually have an extended F pili, meaning MS2 bacteriophage can attack said target through the pili at any time, contributing to the possible effectiveness of MS2 bacteriophages as a vector for antimicrobial effects. Towards the operation of conjugation, F plasmids are true type 4 secretion systems, which is a nanomachine the F plasmid uses to move DNA from a donor cell to a recipient cell (Lawley et al., 2003). The T4S system contains a T4P pilus, which is what MS2 bacteriophages attach to during lysis, and a T4C complex that crosses the periplasm. Later, this is significant towards stoichiometrical modeling of MS2 bacteriophage and ampicillin interactions because this demonstrates a possible necessity for port and metabolic interactions for composite models. For the introduction of T4P and T4S machinery, the gene product “*slt*” locally degrades the peptidoglycan layer in a recipient cell,

allowing the assembly of the T4S complex (Bayer et al., 1995). F pili differs from other bacteria because it is flexible, retracts to bring two cells together, is plasmid encoded versus chromosome encoded, and is mostly used for DNA transfer versus adhesion (Wang et al., 2010).

Persistence, Themes, and Benefits of F Plasmid Targeting

F plasmids are some of the first “superspreader mutations,” dramatically enhancing the conjugational efficiency of conjugative plasmids that belong to diverse incompatibility groups (Virolle et al., 2020). Antibiotic resistance genes are located mainly in an F plasmid; 66% of acquired resistance genes are located in F replicons within *E. coli* (Arismendi et al., 2020). In fact, F plasmids have enhanced transfer capacities compared to IncF plasmids, R100, R6-5, and R1 plasmids branching from the F-class plasmid, using the active FinOP regulatory system (Yoshioka et al., 1987). Lederburg et al.’s 1952 paper on sex compatibility, the discovery of F plasmids and the mechanics of F plasmid spread demonstrates the earliest mention of F plasmid discovery, most notably that the presence of a (then deemed) “F factor” (F⁺ vs F⁻) would mean most bacteria that engaged with this F factor would gain the F⁺ status (Lederburg et al., 1952). >50% of all plasmids can be transferred by conjugation, and most evolutionary models assuming that plasmids must be mobile because plasmid carriage incurs a fitness cost, and the general heterogeneity of male-specific plasmids create the circumstances where male-specific plasmids and mobile plasmids are evolutionarily beneficial despite considerable reproduction drawbacks and fitness costs (Smille et al., 2010; Dionisio, 2005). The cost of antibiotic resistance over time can be substantially distinguished or eliminated by evolution over short periods of time, meaning shortly suspending the use of antibiotics is not an actually viable solution towards antibiotic resistance (Lenski, 1998). However, the usage of MS2 bacteriophages and ampicillin as phage-antibiotic synergetic combinations towards the eradication of bacteria is significant

because it presents the capacity to apply two evolutionary pressures: the pressure to lose the F plasmid or be lysed, and the pressure to keep the F plasmid or be lysed by ampicillin. The usage of conjugation is sufficiently fast to exhibit persistence and is generally applicable to communities containing multiple plasmids and multiple populations (Lopatkin et al., 2017). The speed of conjugation and lysis rate competes with each other, creating a unique circumstance where lysis could lyse all cells containing the F plasmid before the F plasmid persists, a condition that will be monitored in the experiment. After this study on persistence and reversal of plasmid-mediated antibiotic resistance, Lopatkin et al. found that there is a necessity to inhibit the conjugative process for effective resistance reversal. Penttinen et al.'s paper on indirect selection against antibiotic resistance through plasmid-dependent bacteriophages further shows that even if phage-mediated selection is not an effect of immediate eradication, phages can still interfere with the “natural order of things in the microbial realm where evolution towards resistance becomes reality” (Penttinen et al., 2021). One such possibility is the MS2 bacteriophage, MS2 standing for male-specific, meaning this bacteriophage specifically selects and proceeds through the lytic cycle for bacteria exhibiting the F plasmid's pili (Spankie et al., 2020). MS2 bacteriophages infect the bacterium *E. coli* through the F-pilis on *E. coli* bacteriophages, or any bacterium with the F⁺ factor (the F plasmid), by uncoating at any point along the *E. Ccoli* pilis at a distance from the cell body, allowing for the lysis of bacteria (Paine et al., 2025; Cock et al., 2010). After the initial extracellular uncoating, the MS2 bacteriophage uncoats at the cell surface after F-pilis retraction and replicates within the bacteria, ultimately lysing the cell from within. Interestingly, Cock et al. found that MS2 bacteriophages can infect *S. aureus* as well as F⁺ *E. coli*. However, it must be noted that application of bacteriophages to antibiotic resistant bacteria selects for antibiotic sensitive bacteria cells that lose their plasmids

spontaneously and are bacteriophage resistant and thus will end up dominating the bacteria environment. Because the usage of MS2 bacteriophages select for antibiotic sensitive bacteria cells, the literature demonstrates that there is a large possibility for ampicillin - antibiotic synergisms. Bacteriophage resistance can appear in many capacities, mostly emerging in phage absorption prevention, preventing DNA integration by the Superinfection exclusion system, which is when a single protein blocks subsequent infections after an initial infection by blocking infections on various points, degrading phage DNA through restriction-modification systems and clustered regularly interspaced short palindromic repeats (CRISPR), which cut DNA through restriction enzymes and RNA molecules to recognize and target specific DNA sequences (Drulis-Kawa et. al, 2012 & Moghadam et. al, 2020). Bacteria lysins bypass these defences by having a high specificity, cleaving highly conserved structures on bacterium walls, and because lysins are applied externally versus internally, bypassing many traditional resistance methods like active reflux and reduced membrane permeability (Moghadam et. al, 2020 & Wei, Yang, Yu, 2014). Notably, the loss of a receptor(s) on a bacterium are only within inverse gene for gene coevolution patterns, whereas gene for gene coevolutions involve gaining the ability to recognize a portion of a bacteriophage, such as the flagella, and the loss of said portion in the bacteriophage (Burmeister et. al, 2021).

Bacteriophage Synergy

Interestingly, results demonstrated by Cvetanovic et al. showed how *Microvidae* and MS2 bacteriophages can still be successful if used in synergy with existing antibiotic treatments (Cvetanovic et al., 2024). Phages can evolve or co-evolve and consequently restrict the evolution of resistance mechanism in bacteria, synergistic bacterial activity of an isolated phage and ampicillin against urinary tract infecting *Escherichia coli* O157 found that sub-inhibitory

concentrations of ampicillin in mixture with the isolated natural phage was a good alternative to using high doses of antibiotics (Moradpour et al., 2020). This presents a convincing alternative or supplement to traditional usages of antibiotics in agriculture. Lytic bacteriophages in combination with antibiotics show phage-antibiotic synergy in a separate study, where the timing sequence of administering phages and antibiotics can impact the killing effects on different bacterial strains (Liu et al., 2006). The study found that ampicillin had an affinity for PBP2s and PBP3s (penicillin-binding proteins that help construct the cell wall of gram-negative bacteria. In *Salomonella Typhimurium* phage-resistant bacteria also showed signs of reduced virulence and antibiotic resistance, and there was an inverse effect in reference to bacteriophage and antibiotic sensitivities in the targeted bacteria (Laure et al., 2022). This suggests that MS2 bacteriophages can not only evolutionarily select for reduced virulence, the loss of antibiotic resistance overall but also antibiotics can also help contribute to an emphasis of these effects. Phage antibiotic synergy overall is cheaper and more resource efficient in comparison to phage cocktails (Biswas et al., 2024). Interestingly, though overall 2x and 1x minimum inhibitory concentrations (MIC) strains showed no significant difference compared to individual or combination treatments, but at 0.5 MIC values, there was clear PAS (Rastegar et al., 2024). Overall, MS2 bacteriophages are good solutions for the treatment of *E. coli* and other bacteria with the F plasmid gene or F-like plasmid genes because they select against having resistance. This selection allows for a cheaper solution to bacteriophage resistances considering existing materials, follows literature demonstrating that lytic bacteriophages and antibiotics already have synergy, and shows great potential for synergy with the evolutionary forces that ampicillin natively provides.

Digital Modeling and Stoichiometric Constraints

Vivarium is an open source software framework developed by the Covert Lab at Stanford which possesses the ability to build composite models combining many useful modeling frameworks: agent-based models, ordinary differential equations, and stoichiometric reaction systems (Agmon et al., 2022). Vivarium was designed with the goal of integrated whole-cell modeling, but enables interfaces between various ranges of different models in a unified simulation. The nested hierarchies and scalable representation Vivarium was designed with allows for rigid-body physics simulators which can be used to model MS2 bacteriophage particles, ampicillin particles, the movement of bacteria cells, and the interaction of various cellular elements and external environments through modules like bioscrape, genEcoli, and multibody, which can be found on the vivarium github (Vivarium Collective, n.d.). Around that time, Vivarium was used to successfully model cell-cell protein interactions coupled with external ampicillin and heterogenous gene interactions (Skalnik et al., 2021). The flexibility of using Vivarium was that models of antibiotic resistance could be added, and the colony's reaction could be more diversified and simulated. An Agent-Based Model (ABM) is superior to ordinary differential equations and simplified population dynamics through derivative equations because of Alqahtani et al.'s paper on quantification of ampicillin diffusions, being at a rate of 3×10^{-6} to $5 \times 10^{-6} \text{ cm}^2/\text{s}$, yet Armanious et al. showed that the free diffusivity constant of the icosahedral MS2 bacteriophage was 3.3×10^{-8} to $1 \times 10^{-7} \text{ cm}^2/\text{s}$ (Alqahtani et al., 2024; Armanious et al., 2016). These particles move at different rates and individually simulating particles is advantageous because traditional differential equations are deterministic and average out the behavior of a system, often assuming perfectly mixed solutions and infinite populations, but Gillespie's equation captures extrinsic noise and the discrete, random nature of particle collisions (Gillespie, 2000). Despite the age of Gillespie's algorithm, it is still one of two

fundamental approaches to modeling, and since the simplistic nature and modulated nature of Vivarium's multi-CPU based framework allows for fast loading times, the remarked drawback of loading times is not an issue (Albert, 2016; Agmon et al., 2022). Equations treat all *E. coli* in one generalized state, ABMs are superior because it allows for the state of each individual cell, such as F plasmid status, which is the goal of the experiment anyways. Elowitz et al. shows that cells show substantial phenotypical variation, and that intrinsic and extrinsic noise should be factored in (Elowitz et al., 2002). In a separate paper, Gillespie describes the change of chemical populations being by discrete integer amounts, a "random-walk process" because Ordinary Differential Equations (ODE) describe the same result every time, yet Gillespie's Monte Carlo Algorithm allows for the probability of an outcome and inherent fluctuations (Gillespie, 1977). Though Monte-Carlo equations are good for multiple model evaluations using random or pseudo-random numbers (e.g. numbers gathered from a standard deviation range centered at a given constant, usually 1) multiplied to a set array or algorithm, Latin Hypercube Modeling (LHM), introduced by McKay et al. in 1979, allows for an unbiased estimate of the original model's results, requiring fewer samples than simple random sampling (Marino et al., 2009; McKay et al., 1979). To make sure the synergy evaluations and results were robust, LHM is used for global sensitivity analysis to ensure a more uniform coverage of multidimensional input spaces.